Bid-Price Variance & Counterparty Risk Analysis

Analytics Consulting Project · Client: A Leading Construction Firm

*The Problem: Bidding on public infrastructure contracts is highly competitive and largely intuition-driven, making it hard to know when a bid is **priced right, too aggressive, or exposed to high-risk counterparties.***

Project Intro

1. **Data Extraction & Bid-Price Variance**

   Extracted and cleaned 3,500+ public bid records via SQL, then performed bid-price variance analysis to improve bid prioritization.

2. **Counterparty Risk Benchmarking**

   Mapped recurring bid overlap across competitors, identified 28 high-exposure counterparties, and used LOF anomaly detection to flag irregular pricing behavior for enhanced diligence.

3. **Win-Probability & Exposure Modeling**

   Ran a logistic regression model across 4 bid categories to quantify competitor-specific win probabilities, translating project-scale and geographic pricing effects into a 90 bps cut in bid-cost exposure.

4. **Bid Strategy Recommendations**

   Translated model outputs into clear go or no-go signals reviewers could use directly when deciding how aggressively to price a bid.

Impact

$4M+
Potential revenue upside identified

264
Recoverable analyst hours identified

3,500+
Public bid records extracted & modeled

80%
Counterparty match rate vs. watchlist

28
Key counterparties isolated for risk review

90 bps
Cut in bid-cost exposure

Tools & Methods: Python · SQL · KNN Imputation · LOF Outlier Detection · Logistic Regression · Bid-Price Variance & Counterparty Risk Analytics